



HEMANT RAMPHUL

SOFTWARE ENGINEER | EMBEDDED SYSTEMS, AI & ROBOTICS,
15+ years of professional experience.

"Building thoughtful, efficient and elegant solutions to interesting problems."

☎ +33 07 49 92 22 40 ✉ hemantramphul@gmail.com

📍 7, Avenue du Médoc, 33114 Le Barp, Nouvelle-Aquitaine, France



* SKILLS & STRENGTHS

Professional

- Fluent in French, English and Mauritian Creole.
- Effective organisation and prioritisation of workload.
- Quick to adapt to new techniques and technologies.
- Works effectively with minimal supervision.

Personal

- Creative and inventive, able to solve complex problems.
- Effective communicator, reliable and committed to excellence.
- Energetic team player with initiative and attention to detail.

* EDUCATION

■ Master's Degree in Artificial Intelligence and Robotics

University of Mascareignes & University of Limoges
Mauritius & France (Nov. 2021 – Dec. 2024)

■ Bachelor's Degree in Software Engineering

University of Mascareignes & University of Limoges
Pamplemousses, Mauritius (Aug. 2015 – Dec. 2016)

■ Diploma in Information Systems, specialising in Software Development

Swami Dayanand Institute of Management
Pamplemousses, Mauritius (Aug. 2008 – Dec. 2010)

* CERTIFICATIONS

- Microsoft Certified Professional
- Microsoft Certified Solutions Developer : Web Applications
- MCSA : Web Applications
- Android Certified Application Developer

* TECHNICAL SKILLS

■ Programming Languages

- | | | | |
|----------|--------|--------|--------------|
| ▸ C# | ▸ Dart | ▸ Java | ▸ JavaScript |
| ▸ Python | ▸ SQL | ▸ PHP | ▸ TypeScript |

■ Frameworks, CMS & Others (Front-end / Backend)

- | | | | |
|----------------|------------|---------------|------------------|
| ▸ .NET | ▸ Node.js | ▸ WordPress | ▸ Tauri |
| ▸ Flutter | ▸ Xamarin | ▸ Git | ▸ Jupyter |
| ▸ Bootstrap 5 | ▸ Electron | ▸ Agile | ▸ React |
| ▸ Sencha ExtJs | ▸ MATLAB | ▸ GoogleColab | ▸ APIs rest/soap |

■ Software & OS

- | | | |
|------------------------|-----------------|-----------|
| ▸ Android Studio | ▸ VSCode | ▸ Eclipse |
| ▸ Windows/Mac OS/Linux | ▸ Visual Studio | ▸ PyCharm |

■ AI & Robotics

- **Vision** : Image processing, Object detection
- **Robotics** : ROS, Foxglove, Three.js, 3D visualization
- **Video** : WebRTC, RTSP, GStreamer, FFmpeg
- **Telemetry** : MQTT, WebSocket
- **NLP** : Sentiment analysis, NER, Text classification
- **Tools** : TensorFlow, OpenCV, Embedded systems & IoT, PyTorch, Keras, U-Net, YOLO

* PROFESSIONAL EXPERIENCE

Embedded Systems, AI & Robotics Engineer

LYNXDRONE (Sept. 2024 – Present)

- Developed software tools for robots and drones for connectivity, supervision and exploitation of field data.
- Designed applications for robot connectivity, mission tracking and display of operational data.
- Integrated real-time video streaming and robotic telemetry for on-site operation monitoring.
- Built 3D visualizations and point clouds from ROS data to improve analysis of inspected environments.
- Developed robust interface components for control, visualization and exploitation of embedded data, focusing on usability, reliability and maintainability.
- Developed AI solutions for inspection, including image analysis and automated detection of structural anomalies and defects.

Senior Software Engineer

FULLL (Jan. 2024 – Aug. 2024)

- Developed and optimised OCR algorithms to improve result accuracy.
- Backend development with Node.js and cross-platform mobile apps with React Native.
- Maintained web and mobile interfaces, collaborated with the product team, and tracked code via Git and Slack.

Senior Software Engineer

CEGID (Aug. 2011 – Dec. 2023)

- Functional and technical requirements analysis, complexity estimation and development of new features.
- Application maintenance, bug fixing, performance improvements and technical support to the team.
- Applied software security best practices, contributed to UI/UX quality and wrote technical documentation.
- Provided technical support to the team, liaised with architecture to ensure Cegid compliance, and shared technical documentation.
- Agile development and implementation of new features for web and mobile.

PHP Developer

WEB COMPANIES (Feb. 2011 – Aug. 2011)

- Designed and maintained websites and dynamic applications.
- Clean, well-documented PHP development, collaboration with front-end developers and change tracking via Git.

PHP/Drupal Developer

ESOKIA (Aug. 2010 – Jan. 2011)

- Built and maintained Drupal websites with PHP.
- Developed custom Drupal modules and themes to meet client requirements.
- Configured content types, views, taxonomies and user roles.
- Integrated third-party modules and built responsive templates with HTML, CSS and JavaScript.
- Followed team coding standards and documented custom modules.